Projects

From questions to evidence

Methods, results, and limits from modeling, data, and software work.

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Overview

Project archive

Nine projects across modeling, data analysis, and software systems.

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Audio intelligence

VocalPureNet encoding and denoising

Balancing quality, size, and computation.

Method

Time-frequency features and a two-stream U-Net for classification and denoising.

Result

File size fell 25.49%, distortion fell 70.78%, and 14-class accuracy reached about 95%.

Engineering reliability

Thermoelastic optimization for chip packaging

Thermal mismatch and defects in BGA and QFN packaging.

Method

Equivalent material models, finite-element analysis, and sensitivity checks.

Result

Corner displacement 3.82×10^{-2} mm, effective CTE 22.86 ppm/°C, and modulus 27.75 GPa.

Combinatorial optimization

Municipal waste collection routing

Balancing cost and emissions under practical constraints.

Method

Savings, clustering, nearest-neighbor construction, and simulated annealing.

Result

A nine-vehicle plan with station, ten-year cost, and daily carbon estimates.

Cooperative optimization

UAV smoke-screen deployment

Three-dimensional motion, visibility geometry, and coordinated search.

Method

Geometric screening tests with differential evolution, genetic algorithms, and simulated annealing.

Result

One device achieved 4.9 s of screening; three coordinated UAVs reached 16.1 s.

Distributed analytics

MovieLens user preference analysis

Million-rating analysis on a three-node Hadoop cluster.

Pipeline

HDFS and Hive for cleaning, joining, and aggregation.

Findings and limits

Popular films had steadier ratings; the 3% sample limits broader inference.

Machine learning

Titanic survival prediction

A controlled comparison of three classifiers.

Evaluation

Accuracy, confusion matrices, and ROC–AUC on one train/test split.

Result

Random forest performed best with 83.41% accuracy and 0.889 AUC.

Java development

Campus lost-and-found system

An object-oriented command-line information system.

Functions

Accounts, guest access, permissions, and lost-and-found record management.

Structure

Layered services, file persistence, shared validation, and exception handling.

Full-stack · Supabase

Bilingual guestbook and secure admin console

Public messages, administrator replies, and moderation.

Postgres and RLS

Structured message storage with restricted public reads and submissions.

Auth and protection

Supabase Auth, AAL2 TOTP, input checks, a honeypot, and database rate limits.

Applied multivariate analysis

Inference, classification, and dimension reduction for multivariate data.

Methods

Multivariate tests, clustering, discrimination, PCA, and factor analysis.

Interpretation

Compared standardized and raw results to show the effect of scale.